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EM 626 (AI & Machine Learning for Systems)

Final Project Report

For the final assignment, I decided to pursue Project #3 – Bias and Trust in Healthcare AI Summaries. This project topic centers around detecting bias in LLM-generated medical summaries using GNNs trained on knowledge graphs as well as textual indicators (i.e. natural language processing metrics). The final deliverable for this project is a Python script with an end-to-end pipeline that consists of (1) extracting and parsing content from the LLM summaries, (2) building a knowledge graph, (3) inputting feature calculations from the graph into a GNN, (4) detecting overemphasized & underemphasized concepts, and (5) integrating NLP metrics (i.e. coverage, sentiment) for a complete and holistic analysis. The findings from this project should be useful for interpreting the effectiveness of A.I. LLMs when it comes to extracting information from complex articles.

I decided to focus on the medial subject of Alzheimer's disease. I found three separate articles that each focused on a distinct aspect of the Alzheimer's treatment pipeline. The first article (*"Stepwise Approach to Alzheimer's Diagnosis in Primary Care Using Cognitive Screening, Risk Factors, Neuroimaging and Plasma Biomarkers"*) discusses varying factors and conditions that are indicative of a patient having Alzheimer's as well as screening and testing methods. The second article ("*A Review of the Current Status of Disease-Modifying Therapies and Prevention of Alzheimer's Disease"*) outlines current treatments that are given to combat Alzheimer's and cognitive decline. The final article (*"Recent Advances in Alzheimer's Disease: Mechanisms, Clinical Trials and New Drug Development Strategies"*) describes research experiments and clinical examinations that are being conducted to develop medications for Alzheimer's in the near future.

After selecting these three PubMed articles for my project, I designed the following pipeline depicting the steps that needed to be taken for the completion of this assignment:

- Generate structured summaries for each article
    - Prompt ChatGPT to give a concise summary for each article to be exported into a txt file
    - Extract entities and relations from each txt file to create an XML schema for parsing
- Program Script for Knowledge Graph Creation
    - Parse XML content by ingesting and reading content from all three XML files
    - Create nodes and edges from the subject, predicate, and object outlined by each relation
    - Utilize the NetworkX and PyVis libraries in Python to visualize an interactive knowledge graph merging the nodes and edges from all three article summaries
- Analyze Graph Features
    - Calculate the features of node degree and connection density for each node in the knowledge graph
- Use GNN to predict bias labels
    - Assign initial bias labels for each node type based on node degree (i.e. number of connections)
        - Calculate 30th and 70th percentile of all node degrees
        - If node degree is <30th percentile, label node as 0 (underrepresented)
        - If node degree is between the 30th and 70th percentile, label node as 1 (normal)
        - If node degree is >70th percentile, label node as 2 (overrepresented)
    - Convert graph features to PyTorch tensors that return a Data object with three parameters –
        - X: node features
        - Edge-index: graph structure

        - Y: bias labels
    - Create a two-layer Graph Convolutional Network with ReLU activation between layers
        - Layer 1: Collects information from neighboring nodes
        - Layer 2: Further refines node representations
        - Output layer: Re-classifies each node into the 3 bias classes (i.e. underrepresented, normal, overrepresented)
    - Train the GNN Model using Adam optimizer with 0.01 learning rate over 200 epochs
        - Print training loss every 20 epochs to monitor progress
    - Run the GNN model on all nodes and return the predicted bias classes
- Conduct analysis of textual indicators using NLP Metrics
    - Calculate coverage, sentiment, and neutrality of all three article summaries
    - Extract the most-frequent words and bigrams
    - Create a word cloud showcasing the predominant concepts highlighted in the summaries

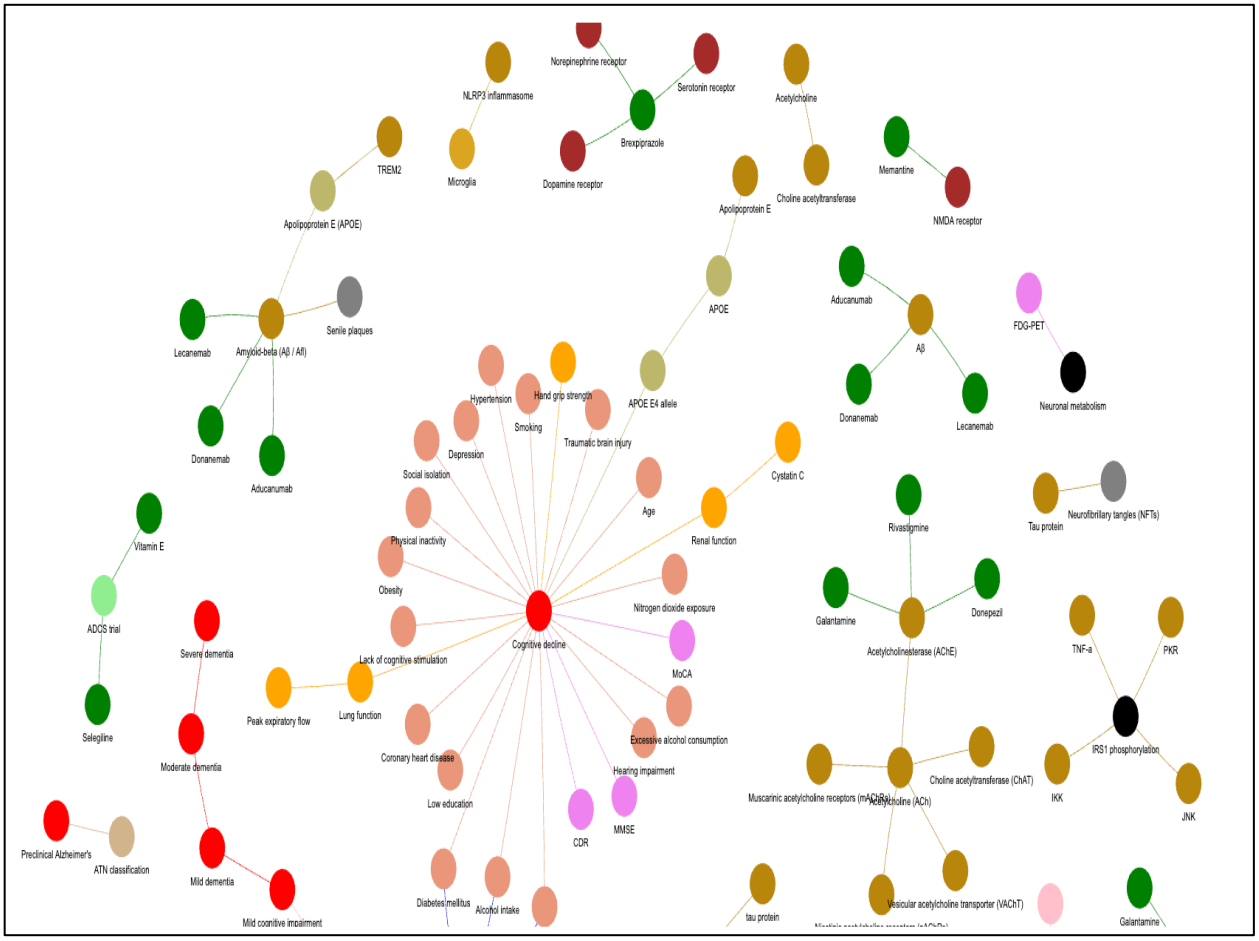
There were a variety of important decisions that needed to be made to ensure that the GNN produced the most insightful results. The first decision was to ensure that "Alzheimer's disease" was not an extracted entity from the article summaries. As I wanted to explore the emphasis on concepts within the umbrella of Alzheimer's diagnosis, treatment, and research, including "Alzheimer's" as an entity would render my analysis ineffective. This would cause all the entities in the knowledge graph to connect to Alzheimer's, and Alzheimer's would be the most-emphasized term with all other topics being under-represented. Therefore, I made sure to not have Alzheimer's or any other related medical conditions present in the extracted content as they would dominate the connection density.

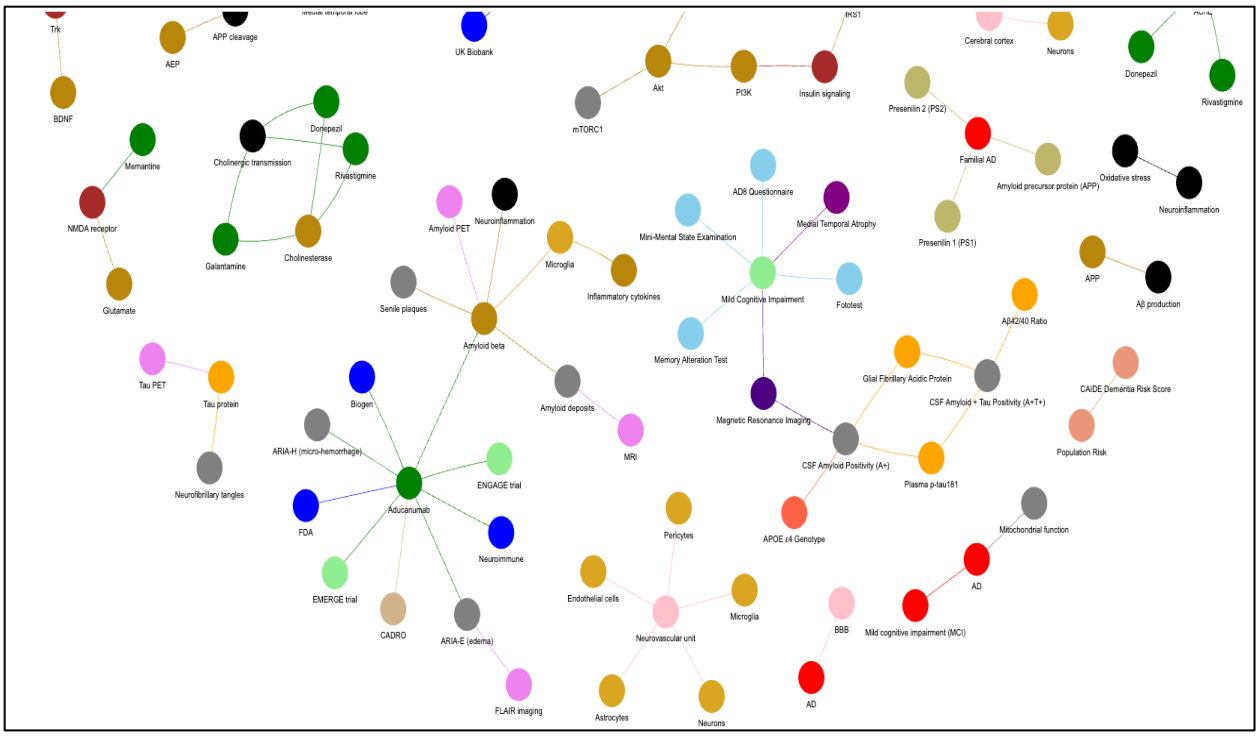
The next key decision was to highlight the node types of the source and target entities forming the extracted relations. This is because node type is the basis of my bias label calculation. I wanted the focus of my GNN analysis to be on what particular over-arching concepts are highlighted (i.e. cells, genes,

proteins, biomarkers). The best way to accomplish this was to have a node-type color scheme in my knowledge graph and a mapping that created ids and encodings for each node's PyTorch tensors based on their type. This would enable me to see which type is most prevalent in the knowledge graph, as well as most frequently over-emphasized from the bias label calculations.

My final key decision was to merge the content from all four txt files summarizing each article for calculating the textual indicators. This would ensure that the NLP metrics corresponded appropriately with the GNN results in terms of what terms/concepts are most-frequently emphasized. The word cloud would be especially insightful, as the sizes of the words would accurately reflect the frequency with which they appear across all three texts.

The first key deliverable of the final project is the knowledge graph created from parsing the AI-generated structured summaries of each article. The knowledge graph is shown below, along with a legend detailing the color-coding for all node types –





Color Key –

- Cognitive Test → sky blue
- Clinical Category, Clinical Trial → light-green
- Biomarker → orange
- Pathology, Pathway, Pathological Feature → gray
  Imaging → indigo
- MRI Feature → purple
- Test → violet
- Genetic Factor → tomato red
- Risk Score, Risk Factor → dark salmon
- Disease, Condition → red
- Drug → green
- Organization → royal blue
- Gene → khaki
- Cell → gold
- Protein → dark gold
- Molecular Process → black
- Receptor, Signaling Pathway → brown
- Structural Factor → pink

The center of the knowledge graph shows a ring of risk factors encircling the condition “cognitive decline” (shown in red). Some other interesting observations are that there is a large presence of green nodes, indicating an emphasis on drugs and medications discussed across the summaries. There is also an abundance of golden nodes, implying a large frequency of cells and proteins indicative of Alzheimer’s mentioned in the texts. Overall, the most emphasized node types from the knowledge graph appear to be risk factors, drugs, cells, proteins, and pathological factors.

The next key deliverable for this project is the knowledge graph feature calculations. I have attached a snapshot of these statistics below –

node	name	type	in_degree	out_degree	total_degree	local_density	global_density
e110	Mild cognitive impairment	Disease_Stage	1	1	2	0.6666666666666670	0.005357142857142860
e111	Preclinical Alzheimer's	Disease_Stage	1	0	1	1.0	0.005357142857142860
e112	Lewy body dementia	Disease	0	0	0	0.0	0.005357142857142860
e113	Vascular dementia	Disease	0	0	0	0.0	0.005357142857142860
e114	Frontotemporal dementia	Disease	0	0	0	0.0	0.005357142857142860
e115	Stroke	Condition	0	0	0	0.0	0.005357142857142860
e116	Chronic traumatic encephalopathy	Condition	0	0	0	0.0	0.005357142857142860
e117	Cognitive decline	Condition	24	0	24	0.08	0.005357142857142860
e118	Severe dementia	Disease_Stage	1	0	1	1.0	0.005357142857142860
e119	Moderate dementia	Disease_Stage	1	1	2	0.6666666666666670	0.005357142857142860
e120	Mild dementia	Disease_Stage	1	1	2	0.6666666666666670	0.005357142857142860
e121	Amyloid beta 42	Biomarker	0	0	0	0.0	0.005357142857142860
e122	Tau protein	Biomarker	1	1	2	0.6666666666666670	0.005357142857142860
e123	Amyloid beta oligomers	Biomarker	0	0	0	0.0	0.005357142857142860
e124	Synaptic markers	Biomarker	0	0	0	0.0	0.005357142857142860
e125	Cystatin C	Biomarker	1	0	1	1.0	0.005357142857142860
e126	Senile plaques	Pathological_Feature	1	0	1	1.0	0.005357142857142860
e127	Neurofibrillary tangles	Pathological_Feature	1	0	1	1.0	0.005357142857142860
e128	Amyloid deposits	Pathological_Feature	2	0	2	0.6666666666666670	0.005357142857142860
e129	ARIA-E (edema)	Pathological_Feature	2	0	2	0.6666666666666670	0.005357142857142860

As expected, the condition node “cognitive decline” has the most in-degrees (24). Another intriguing observation is that the node types Test, Cognitive Test, and Clinical Trial had the highest average local density (1), followed by the node type Risk Factor [0.941 → 14/17(1) + 3/17(0.67)], implying that these are the most tight-knit clusters out of any node group. In terms of total node degrees, the node types with the highest count of this value were Protein (56) and Drug (34). This corroborates what was seen in the knowledge graph, and reinforces that these two subjects are the most

frequently discussed in the summaries. Finally, the global density of the entire knowledge graph is 0.005, indicating that there is not really any inter-connectedness between the nodes and clusters.

The next output produced by the Python Script is the initial bias labels for the nodes. I have attached a snapshot of these values below –

node	name	bias_label
e110	Mild cognitive impairment	2
e111	Preclinical Alzheimer's	1
e117	Cognitive decline	2
e118	Severe dementia	1
e119	Moderate dementia	2
e120	Mild dementia	2
e122	Tau protein	2
e125	Cystatin C	1
e126	Senile plaques	1
e127	Neurofibrillary tangles	1
e128	Amyloid deposits	2
e129	ARIA-E (edema)	2
e130	ARIA-H (micro-hemorrhage)	1
e131	MRI	1
e132	FDG-PET	1
e133	Amyloid PET	1
e134	Tau PET	1
e135	MMSE	1
e136	MoCA	1
e137	CDR	1
e138	FLAIR imaging	1

Scanning the bias labels, the nodes with the highest proportion of “2” as their label are those of the node type Disease and Condition (e.g. mild cognitive impairment, cognitive decline, severe/moderate dementia). Other node types with a high quantity of “2” labels are proteins (e.g. Tau Protein, APOE), drugs (e.g. Aducanumab, Donepezil), and cells (neurons, microglia). This means that these node types have the highest bias and are the most over-emphasized across the texts. All the risk factor nodes (e.g. air pollution, alcohol intake, obesity, smoking, depression) have “1” labels, as do a high

proportion of the test and imaging nodes (e.g. AD8 questionnaire, memory alteration test, FLAIR imaging).

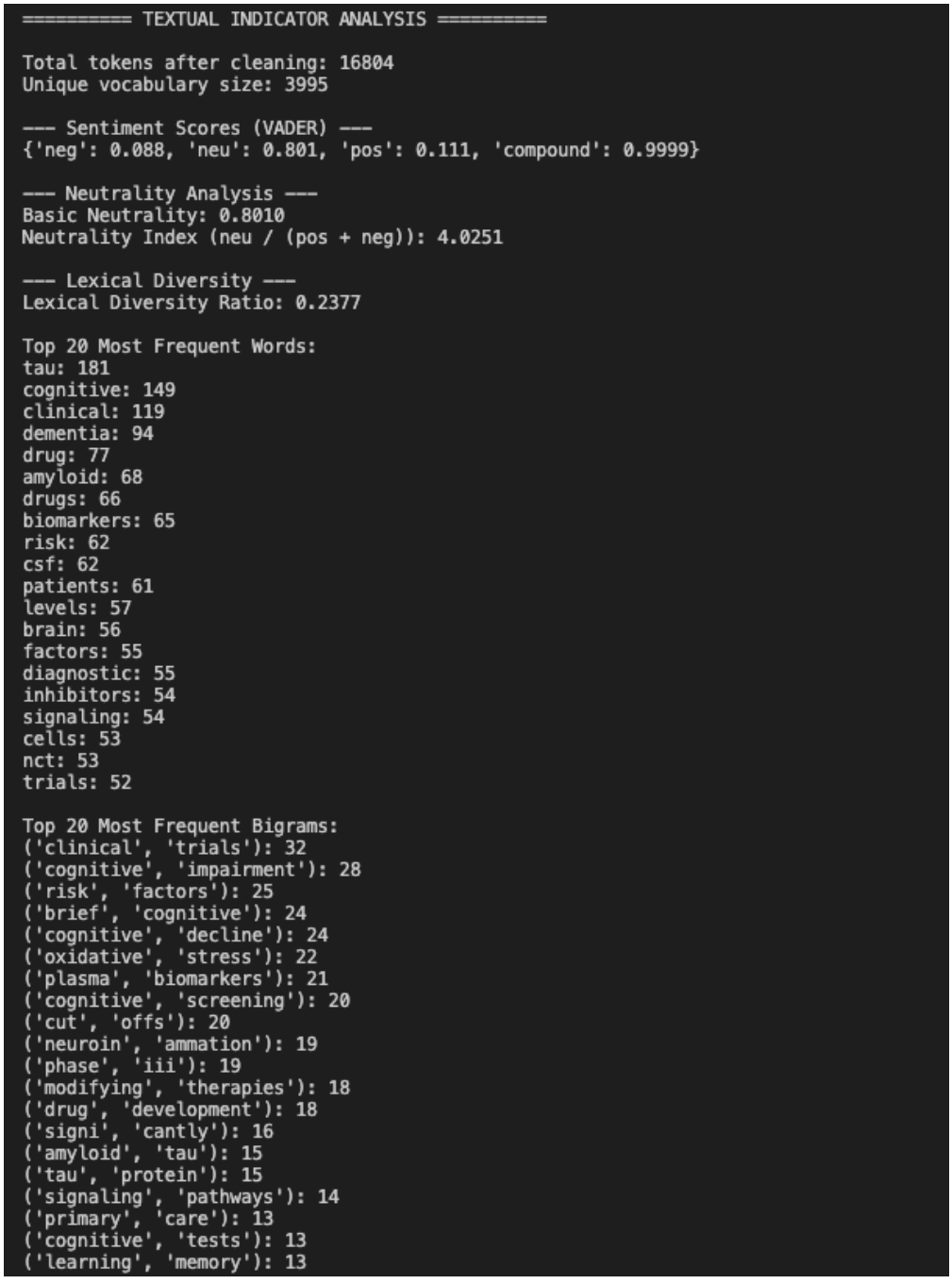
Here are the bias level predictions for each node after GNN training:

node_id	name	type	true_bias_label	predicted_bias_label	prob_underrepresented	prob_normal	prob_overemphasized
e110	Mild cognitive impairment	Disease_Stage	2	2	4.11231354524588E-10	0.0020532491616904700	0.9979467988014220
e111	Preclinical Alzheimer's	Disease_Stage	1	1	2.98655856989516E-11	0.9999958276748660	4.20283731727977E-06
e117	Cognitive decline	Condition	2	2	3.51595958125948E-38	2.31433521236113E-08	1.0
e118	Severe dementia	Disease_Stage	1	1	9.96531052122052E-11	0.9939146637916570	0.0060852752067148700
e119	Moderate dementia	Disease_Stage	2	2	9.63752203042567E-11	0.0069312662817537800	0.9930688142776490
e120	Mild dementia	Disease_Stage	2	2	5.79257752875151E-11	9.27624714677222E-05	0.9999072551727300
e122	Tau protein	Biomarker	2	2	1.38692720502709E-11	0.0011310449335724100	0.9988688826560970
e125	Cystatin C	Biomarker	1	1	1.0990651444498E-10	0.992882251739502	0.007117713801562790
e126	Senile plaques	Pathological_Feature	1	1	4.59472598721622E-16	0.9999996423721310	3.89868915817715E-07
e127	Neurofibrillary tangles	Pathological_Feature	1	1	6.94774723863922E-11	0.9999769926071170	2.30017485591816E-05
e128	Amyloid deposits	Pathological_Feature	2	2	4.54284018993877E-17	1.94871944358965E-07	0.9999997615814210
e129	ARIA-E (edema)	Pathological_Feature	2	2	5.34674777817259E-16	1.76862386069843E-06	0.9999982118606570
e130	ARIA-H (micro-hemorrhage)	Pathological_Feature	1	1	3.50760748029332E-15	0.9999979734420780	2.03203649107309E-06
e131	MRI	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300
e132	FDG-PET	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300
e133	Amyloid PET	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300
e134	Tau PET	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300
e135	MMSE	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300
e136	MoCA	Test	1	1	6.18143869424159E-11	0.9997965693473820	0.00020346442761365300

The GNN training does not affect the bias level values for any of the nodes. This is because it is being trained on the assigned bias labels, which act as a "ground truth" dataset. The GNN takes the initial bias labels and learns how to predict bias labels for other nodes based on their graph features (e.g. node in-degrees, node out-degrees, node type, local density). After training finished, the GNN then runs back through the nodes from the Alzheimer's knowledge graph. Overall, the GNN results support the findings uncovered from the knowledge graph and graph features: drugs, proteins, cells, and related conditions are the most over-represented in the article summaries.

The final piece of this assignment is conducting textual analysis based on NLP indicators. The values calculated are attached below. The first metric is neutrality, which was determined to be 0.8. This means the overall text from the generated texts is neutral in tone, which is to be expected as they are summarizing informative research articles. The next metric, showcasing the coverage of the texts, is the Lexical Diversity Ratio. This value is 0.2377, indicating a very low amount of unique words. This is also to be expected, as the GNN results and graph features show that there is a large amount of over-

emphasized terms. To showcase what these terms are, I decided to extract the most frequent words and bigrams. The most common words were tau (i.e. a protein), cognitive, clinical, dementia, drug, amyloid (also a protein), and biomarkers. The most common bigrams are ‘clinical trials,’ ‘cognitive impairment,’ ‘risk factors,’ ‘brief cognitive,’ ‘cognitive decline,’ and ‘oxidative stress.’



The expected results from the most frequent words and bigrams are the ‘tau’ / ‘tau protein’, ‘drug’ / ‘drug development’, ‘amyloid’ / ‘amyloid tau’, ‘cognitive’ / ‘cognitive impairment’ / ‘cognitive decline’, ‘dementia’, ‘biomarkers’ / ‘plasma biomarkers’, and ‘inhibitors.’ These correspond with the node types (i.e. proteins, drugs, conditions, cells, biomarkers) that had high bias values in the GNN. However, there are also some unexpected results. These include ‘clinical’ / ‘clinical trials’, ‘risk’ / ‘risk factors’, ‘diagnostic’, ‘oxidative stress’, ‘cognitive screening’ / ‘cognitive tests’, and ‘learning memory.’ These terms are unexpected as they correspond to node types (i.e. clinical trials, tests, risk factors) that had average bias labels. For these to be amongst the most-frequent terms therefore means that the GNN judged these subjects to be important and that they were not over-emphasized.

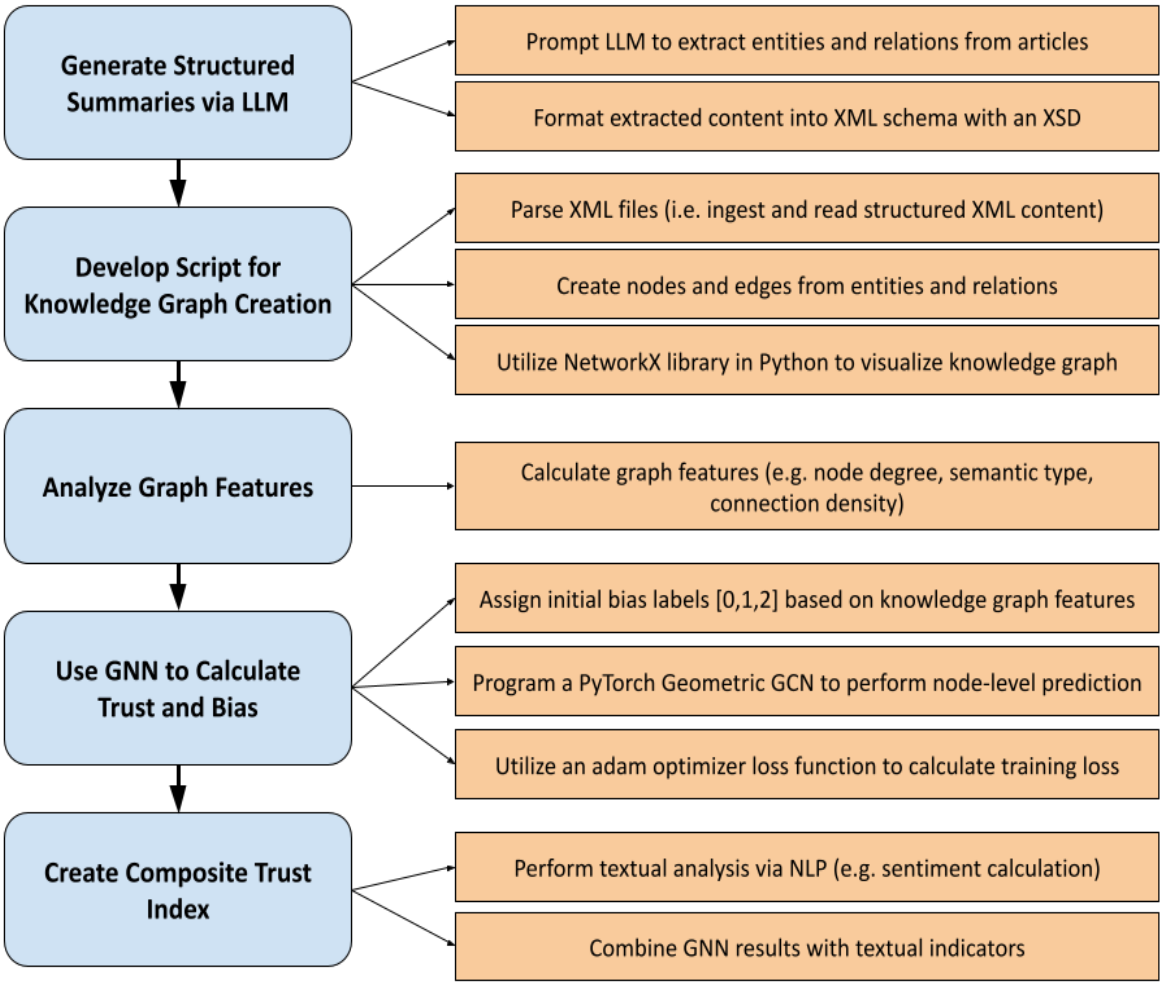
Bringing together the results of both the GNN bias labels and textual indicators yields some key insights towards the effectiveness of AI-generated summaries. Specifically for the three articles I utilized related to Alzheimer’s disease, there seems to be a high bias for the LLM (i.e. ChatGPT) to focus on drug developments, diseases and conditions related to Alzheimer’s, and cells and proteins that are indicative of cognitive decline. These concepts were over-represented in the summaries when they were not as pre-dominant in the articles. On the other hand, important topics such as risk factors that cause Alzheimer’s as well as clinical trials and tests utilized for diagnosis were represented a fair amount in the summaries. Branching out from the context of Alzheimer’s, these results indicate that AI-generated summaries tend to have a high bias towards technical subject matter and details. This could have adverse effects on users as people utilizing this generated content will be presented with an over-emphasis on jargon and potentially miss out on broader, more important concepts.

Appendix

Prompts Used →

- Please generate a concise summary of the following research articles related to Alzheimer's diagnosis, treatment, and drug development.
- Extract entities and relationships from the summaries and format these into an XML file.
- Develop a simple GNN that can predict bias labels based on the knowledge graph features of node in-degrees, node out-degrees, node type, local density, and global density. This GNN should be trained on the nodes in the Alzheimer's knowledge graph.

Pipeline Diagram →



Articles Used →

Altuna, Miren, et al. “Stepwise Approach to Alzheimer’s Disease Diagnosis in Primary Care Using Cognitive Screening, Risk Factors, Neuroimaging and Plasma Biomarkers.” Nature News, Nature Publishing Group, 27 Aug. 2025, https://www.nature.com/articles/s41598-025-17394-3

Parums, Dinah V. “A Review of the Current Status of Disease-Modifying Therapies and Prevention of Alzheimer’s Disease.” Medical Science Monitor : International Medical Journal of Experimental and Clinical Research, U.S. National Library of Medicine, 13 May 2024, https://pmc.ncbi.nlm.nih.gov/articles/PMC11097689/

Zhang, Jifa, et al. “Recent Advances in Alzheimer’s Disease: Mechanisms, Clinical Trials and New Drug Development Strategies.” Nature News, Nature Publishing Group, 23 Aug. 2024, https://www.nature.com/articles/s41392-024-01911-3#Abs1